Stress Test

Boolean Access Control Optimization Challenge

Imagine you're just joined a cybersecurity team working on programming an Access Control System using the C language. A junior programmer wrote a function that checks whether a user is allowed to access the sensitive control panel or not.

The basic conditions depend on three variables:

A : The user is an admin (is_admin)

M : The user is a manager (is_manager)

S : The connection is encrypted and secure (is_secure)

The junior programmer wrote the follwing condition in the code:

```
if((is_admin && !is_manager && !is_secure) ||
  (is_admin && !is_manager && is_secure) ||
  (!is_admin && is_manager && is_secure) ||
  (is_admin && is_manager && is_secure))
{
  unlock_control_panel();
}
```

As the system engineer reviewing this code (Code Review), you immediatly realize that this long condition wastes unnecessary clock cycles and translates into an excessively large number of logic gates at the hardware level.

The Engineer Challenge (The Stress Test)

Your task is to "reverse engineer" this condition and simplify it as much as possible.

Solveing The Problem:

The original Boolean equation
(using Bolean Algebra)

$$A\bar{M}\bar{S} + A\bar{M}S + \bar{A}MS + AMS$$

simplify:

$$MS(\bar{A}+A) + A\bar{M}S + A\bar{M}\bar{S}$$

$\bar{A}+A=1$

$$\Rightarrow MS + A\bar{M}S + A\bar{M}\bar{S}$$

$$A\bar{M}(S+\bar{S}) \Rightarrow A\bar{M}$$

Finally: $MS + A\bar{M}$

The new code:

```
if ( (is_admin && !is_manager) ||
     (is_manager && is_secure) )
  {
   unlock_control_panel();
  }
```

Why the new code is more effective?

The new code is more effective because it performs the same access check using fewer logical operations, which reduces computation and makes the code easier to read and maintain.

E-Vex - Day 10